UPDATA · ESP32 PHONE BRIDGE

ESP32 to CRM
Setup without source code

Mac • Windows PC • Linux | 13 November 2026

Use this guide to prepare an original ESP32, connect an Android SIM phone and verify real calling audio. End users do not need access to GitHub, the CRM repository, firmware source or ESP-IDF.

> Two separate downloads in CRM Setup guide: **Download gateway** installs the computer program; **Download ESP32 firmware** provides compiled images for a new board. Neither download flashes a board automatically.

Connection: CRM ↔ internet ↔ computer gateway ↔ USB ↔ ESP32 ↔ Bluetooth hands-free ↔ Android SIM.

Contents: 1 Equipment; 2 Gateway; 3 USB checks; 4 Flashing tools; 5 Chip checks and backup; 6 Flash compiled firmware; 7 CRM link; 8 Bluetooth and sound; 9 Daily use and moving computers; 10 Troubleshooting and release information.

Use current files from the same approved CRM release. Project physical verification used macOS; Windows/Linux instructions still require real hardware and audio acceptance. Stop at an unexplained failure.

1. Equipment and preparation

1. Use an original ESP32-WROOM-32 / 32E / 32D with Classic Bluetooth. S2, S3, C3, C6 and H2 do not support this bridge. CP210x/CH34x identifies the USB adapter, not the ESP32 silicon.
2. Use a USB data cable, direct computer port, Android phone with an active SIM, calling credit and signal. Have a second phone you own available for testing.
3. Install an actively supported Node.js LTS version, 20 or newer, from nodejs.org/en/download. Choose the correct computer architecture. Reopen Terminal afterward.
4. Sign into CRM with AI Calling write permission or administrator access. Use your own phone card.
5. Before diagnostics or flashing, finish calls, stop gateway processes and any Mac login service, and close screen / Arduino Serial Monitor.

```
node --version
npm --version
```

For an already prepared board, try the USB check first. Normal reconnection does not need firmware flashing, BOOT or a new security key.

Replace placeholders such as PHONE_ID and YOUR_CODE with values from the phone card. Use actual ports from discovery; Mac /dev/cu.usbserial-0001, Windows COM5 and Linux /dev/ttyUSB0 below are examples. Do not paste shell prompts or backslashes inside https://.

2. Install the computer gateway

1. Open AI Calling → Connections / Paired phones → Setup guide. Download gateway.
2. Extract the archive into a permanent folder you own. Open the package folder; it contains bin, src and package.json.
3. Mac: open Terminal, type cd followed by a space, drag the package folder from Finder and press Enter. Windows: open cmd.exe, type cd /d followed by the quoted package path. Linux: open a terminal in the package folder.
4. Run the installation command below and wait for it to finish.

```
npm ci --omit=dev
node bin/esp32-probe.mjs ports
```

Keep this folder and Node installation in place. Each phone needs its own prepared bridge, USB port, profile and gateway process.

> This installs the gateway only. Do not search for firmware source in this package. A separate compiled firmware download is available for new boards.

3. Verify the USB connection

With the gateway stopped, list ports and use the actual bridge port:

```
node bin/esp32-probe.mjs ports
```

Mac example:

```
node bin/esp32-probe.mjs info --port /dev/cu.usbserial-0001
```

Windows example:

```
node bin/esp32-probe.mjs info --port COM5
```

Linux example:

```
node bin/esp32-probe.mjs info --port /dev/ttyUSB0
```

1. Firmware information appears: USB handshake works; skip flashing and continue at section 7.
2. Resource busy: identify and close the program holding the port. On Mac use lsof as below. Do not run the gateway and probe together.
3. No bridge listed: try a known data cable, direct USB port and the adapter manufacturer’s driver.
4. [no_reply]: release BOOT, press EN/RESET once, wait five seconds, retry. If it still fails, inspect the bootloader and firmware using sections 4–6. A timeout alone does not prove reflashing is necessary.

```
lsof /dev/cu.usbserial-0001 /dev/tty.usbserial-0001
```

A power LED, Android Bluetooth name or online CRM gateway does not prove the ESP32 answered over USB.

4. Get compiled firmware and flashing tools

1. In CRM Setup guide, expand New board firmware. Choose Download ESP32 firmware. If unavailable, ask the administrator to include the binary release; repository access is not needed.
2. Extract updata-esp32-firmware.tgz. The updata-esp32-firmware folder contains bootloader.bin, partition-table.bin, updata_hfp_bridge.bin, manifest.json, flash_args and INSTALL.txt. No source checkout is needed.
3. Install Python 3.12 from python.org/downloads. Create an isolated tool environment below; this does not build firmware.

Mac / Linux Terminal:

```
python3.12 -m venv "$HOME/.updata/flash-tools"
source "$HOME/.updata/flash-tools/bin/activate"
```

Windows Command Prompt (cmd.exe):

```
py -3.12 -m venv "%USERPROFILE%\.updata\flash-tools"
call "%USERPROFILE%\.updata\flash-tools\Scripts\activate.bat"
```

Then on either platform:

```
python -m pip install esptool==4.12.0
python -m esptool version
```

Activate the environment again in every new flashing command window. On Linux, ensure your user has USB serial permission; on Ubuntu this commonly means dialout membership and a new login. Do not run the gateway as root.

5. Inspect and back up the actual board

Stop gateways and serial tools. Set the actual port in your activated tool window:

Mac/Linux (change the value for Linux):

```
export UPDATA_ESP_PORT="/dev/cu.usbserial-0001"
```

Windows cmd.exe:

```
set "UPDATA_ESP_PORT=COM5"
```

In every command below, replace PORT with the actual port. Do not type PORT literally.

```
python -m esptool --chip esp32 --port PORT chip_id
python -m esptool --chip esp32 --port PORT flash_id
python -m espefuse --chip esp32 --port PORT summary
```

> Continue only for an unsecured original ESP32 with 4 MB flash. Stop for secure boot, flash encryption, different size/chip or unclear results. Do not change eFuses, force flags or erase-flash.

If Connecting stalls, hold BOOT, press/release EN and release BOOT once it connects. For normal running, keep BOOT released.

Make a private backup folder and use a fresh filename. Mac/Linux:

```
mkdir -p "$HOME/updata-esp-backups"
chmod 700 "$HOME/updata-esp-backups"
umask 077
```

Windows: create a private folder under your user profile and restrict its access. Replace BACKUP.bin below with a new absolute file path in that folder.

```
python -m esptool --chip esp32 --port PORT --baud 460800 --after no_reset read_flash 0 0x400000 BACKUP.bin
python -m esptool --chip esp32 --port PORT --baud 460800 --before no_reset --after no_reset verify_flash 0
BACKUP.bin
```

Run verify immediately, without reset/unplug/application startup. Stop unless it succeeds. Backups exclude eFuses and RAM; keep them private.

6. Flash the downloaded binary release

1. Open your activated flashing command window in the extracted updata-esp32-firmware folder. On Mac type cd plus a space, drag the folder into Terminal and press Enter. Windows uses cd /d plus the quoted folder path.
2. Read manifest.json: check release version, chip, 4 MB size and expected images. The archive is generated from an approved build and its file hashes are checked by the release packager. Do not mix versions or change flash offsets.
3. After the chip/security and verified backup checks, run the command below using the actual port.

```
python -m esptool --chip esp32 --port PORT --baud 460800 write_flash @flash_args
```

The included flash_args selects DIO / 40 MHz / 4 MB with bootloader at 0x1000, partition table at 0x8000 and application at 0x10000. Updating another partition layout may replace saved settings.

1. Wait for successful flash completion. Release BOOT, press EN once and wait five seconds.
2. Return to the gateway package folder. Use its info command with your actual port.

```
node bin/esp32-probe.mjs info --port PORT
```

Checkpoint: firmware and board information must appear. If no reply remains, collect the complete flash output for support. Do not repeatedly flash or change the CRM key. The running binary protocol uses 921600 baud automatically; flashing uses 460800.

7. Link the computer to CRM

1. For a new phone: Add paired phone → name/SIM label → Create pairing code. For an existing card: stop its old gateway/service and choose Link computer on that same card. Do not create a duplicate.
2. Copy the generated pairing command, including actual API address, one-use code and PHONE_ID. Codes expire after ten minutes; a replacement code keeps the same card.
3. Run the generated command in the installed gateway package. Add the actual --port if necessary, especially with several bridges.

Example only; replace values with the phone card's generated values:

```
node bin/updata-gateway.mjs pair --api "https://YOUR_CRM_HOST/api/v1" --code YOUR_CODE --profile PHONE_ID --port PORT
node bin/updata-gateway.mjs status --profile PHONE_ID --check-crm
```

After paired:true and a successful CRM check, start:

```
node bin/updata-gateway.mjs run --profile PHONE_ID
```

If the saved USB port changed:

```
node bin/updata-gateway.mjs run --profile PHONE_ID --port PORT
```

Keep Terminal open. Computer must report online and ESP32 USB connected. status without --profile checks legacy/default credentials, which may differ. The frontend website address and API address can differ; use the generated command.

8. Security key, Android and sound

1. Open Phone setup. If the live board and CRM already report a matching provisioned key, keep it. If pending, hold BOOT while choosing Save security key; release after success. EN is reset.
2. Choose Pair a new phone for a two-minute discovery window. On Android open Bluetooth → Pair new device and select the exact Updata Phone Bridge name on the card. Confirm the pairing number if prompted.
3. Enable Phone calls for that Bluetooth device. Select Android's default calling SIM, rather than Ask every time. Disconnect competing headsets/Mac hands-free connections if they take audio.
4. Choose Connect phone in CRM. For normal reconnection to an already bonded phone, try Connect phone before pairing again.
5. Open Sound & test call. Enter your own second phone's international number. Confirm ownership and authorize one test call. Answer on that second phone.
6. Listen to the AI and say its requested phrase. Confirm the AI understands your response. Adjust AI volume if needed.
7. Choose Verify call audio only after speech works both ways. Confirm both checks and save. Repeat after a firmware/host change, even if an old check is green.

> Acceptance: current Computer online, USB connected, security key provisioned on the board, phone HFP connected, two-way audio verified and readiness OK.

If the test reports an unavailable AI agent, the administrator must check the voice worker and LiveKit connection. Replacing the ESP32 key does not fix a server voice-service error.

9. Daily use and a different computer

Keep the computer awake and online, ESP32 connected and Android Bluetooth on. Start the gateway with its existing profile. No routine reflash or new key is required.

Mac login startup: after foreground calls work, stop with Ctrl+C and run from the permanent package:

```
node bin/updata-gateway.mjs service install --profile PHONE_ID
node bin/updata-gateway.mjs service status --profile PHONE_ID
```

Before diagnostics or moving hosts, stop/remove that service:

```
node bin/updata-gateway.mjs service uninstall --profile PHONE_ID
```

Windows/Linux use the foreground run command; the built-in login installer supports macOS only.

1. Move an existing bridge/phone: finish calls, stop the old gateway/service, connect USB on the new computer, install gateway and discover its new port.
2. If info returns Updata firmware, do not flash again. Use Link computer on the same CRM card with a fresh code, that phone's profile and new port.
3. Keep a matching key and Bluetooth bond; reconnect and test real sound on the new host.
4. For another phone, use another prepared ESP32, distinct port/profile and separate gateway process. Do not open one bridge twice.

Credentials stay in the computer user's .updata/profiles/PHONE_ID folder. Link through CRM on the new computer; do not share credential files, tokens or keys.

10. Troubleshooting and handover

Resource busy: close screen, serial monitors, other gateways and login services. Identify the actual port owner; do not kill unrelated programs.

No reply: release BOOT, reset EN once, retry. Check cable/port, bootloader response and approved firmware. A remembered Bluetooth entry does not verify USB.

Not linked: check the correct profile/computer user. Use Link computer on the existing card and a new code. Do not rotate the calling key for missing computer credentials.

Computer online, USB unreachable: CRM heartbeat is working, hardware control is not. Free/reconnect USB and follow section 3.

Firmware download missing: ask the administrator to publish the compiled binary release. End users do not need repository access.

Bluetooth missing: scan during Pair a new phone, enable Phone calls, choose default SIM and reconnect. Check current CRM status.

Failed AI/audio test: check live USB, HFP, key, SIM and voice-worker readiness. Do not mark speech verified based on power or frame counts alone.

Record: computer/Node version, gateway folder/version, firmware version, board identity/size, backup location, actual port, API URL, profile ID and real audio result. Share errors/logs with secrets and personal numbers removed.

Release: this guide accompanies compiled firmware 0.1.4, original ESP32, ESP-IDF v5.5.6 build, DIO / 40 MHz / 4 MB. The firmware bundle contains only three binary images, manifest, flash arguments and instructions; no CRM/firmware source, provisioned keys, NVS snapshots or backups.

Official tools: Node.js · Python · Espressif esptool commands · CP210x driver

> This guide does not deploy CRM services or prove a specific physical board passed. Complete the live acceptance checks. Never use erase-flash or change security eFuses as a general repair.